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Condensation reaction using monohydroxymethylated cardanol for cardanol–phenol bisphenol with ultra-low hormone receptor binding affinity

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A novel bio-based cardanol–phenol bisphenol, analogous to bisphenols A/F, with hormone receptor binding affinity nearly three orders of magnitude lower than that of bisphenol A, was designed, synthesized, and characterized. The synthetic strategy capitalizes on the differential reactivity at the *ortho* and *para* positions of phenol, induced by the cardanol side chain.

Bisphenols are the basic blocks of various polymers, such as polycarbonate and epoxy resins with an annual global output exceeding 10 million tons.[1–5] The most widely used bisphenols are bisphenol A (BPA) and bisphenol F (BPF), which account for over 85% of global bisphenol production. However, due to their structural similarities to sex hormones, both BPA and BPF can be bound to estrogen receptors (ER, such as ERα and ERβ, SI part 1), which can activate estrogenic signaling pathways leading to precocious puberty or disruption of the expression of androgen responsive genes.[6–11] Consequently, both BPA and BPF exhibit strong hormone receptor binding affinity, posing significant long-term risks to human health and the ecosystem.[12]

The strong binding affinities of BPA and BPF to hormone receptors are attributed to their comparable molecular sizes and the high polar –OH groups, which can form hydrogen bonding with hormone receptors (SI part 1).[13–15] Tailoring the molecular structure of bisphenols, including the size and the hydrophilicity, are key strategies to reduce their affinities to hormone receptors.[16] Recent studies indicate that BPA/BPF derivatives retain strong hormone receptor binding affinity, as simple modifications do not sufficiently alter the structural similarity between bisphenols and hormone molecules.[12,17–20] Chen *et al.* found that many BPA/BPF analogs such as BPAF, BPB, and BPS exhibited estrogenic and/or antiandrogenic activities similar to or even greater than that of BPA (molecular structure of BPAF, BPB, and BPS can be found in Fig. S3).[21] Practically, the European Union (EU) has announced BPA ban (EU)2024/3190 to prohibit the use of BPA, other bisphenols, and their derivatives in food fields. Seeking alternatives to BPA/BPF with low hormone receptor binding affinity is crucial for the economy, ecology, and human health.

Bisphenols are generally prepared by the condensation reaction of phenol and acetone (for BPA) or formaldehyde (for BPF) with a large excess of phenol under strongly acidic conditions (SI part 2).[22–24] Cardanol is a phenol substitute extracted from cashew nut shell liquid, and different from phenol, it has a long unsaturated C_{15} side chain.[25–27] As shown in Fig. 1a, when one phenol in bisphenols A/F was replaced with cardanol, a novel cardanol–phenol bisphenol with a larger molecular size and less hydrophilicity compared to BPA/BPF can be prepared, which may solve the hormone receptor binding affinity issues of bisphenols. However, in the condensation reactions for bisphenol preparations, large excessive phenols are needed to avoid the formation of phenolic resin, and strong acidic catalysts are required.[28,29] Due to the higher reactivity of phenol than cardanol and the high reactivity of C=C bonds under acidic conditions, cardanol–phenol bisphenol cannot be synthesized by tailoring the molar ratio of the raw materials. Until now, cardanol–phenol bisphenol has only been reported by phenolization of the aliphatic chain of cardanol (NC-514 from Cardolite Corporation).[28,30] However, due to the polyene nature of cardanol (41% cardanol triene, 22% cardanol diene, 34% cardanol monoene, and 3% saturated olefin), the monotonicity of these bisphenols is limited. Bisphenols with free flexible chains *via* condensation of cardanol and phenol analogs to BPA/BPF have never been reported.

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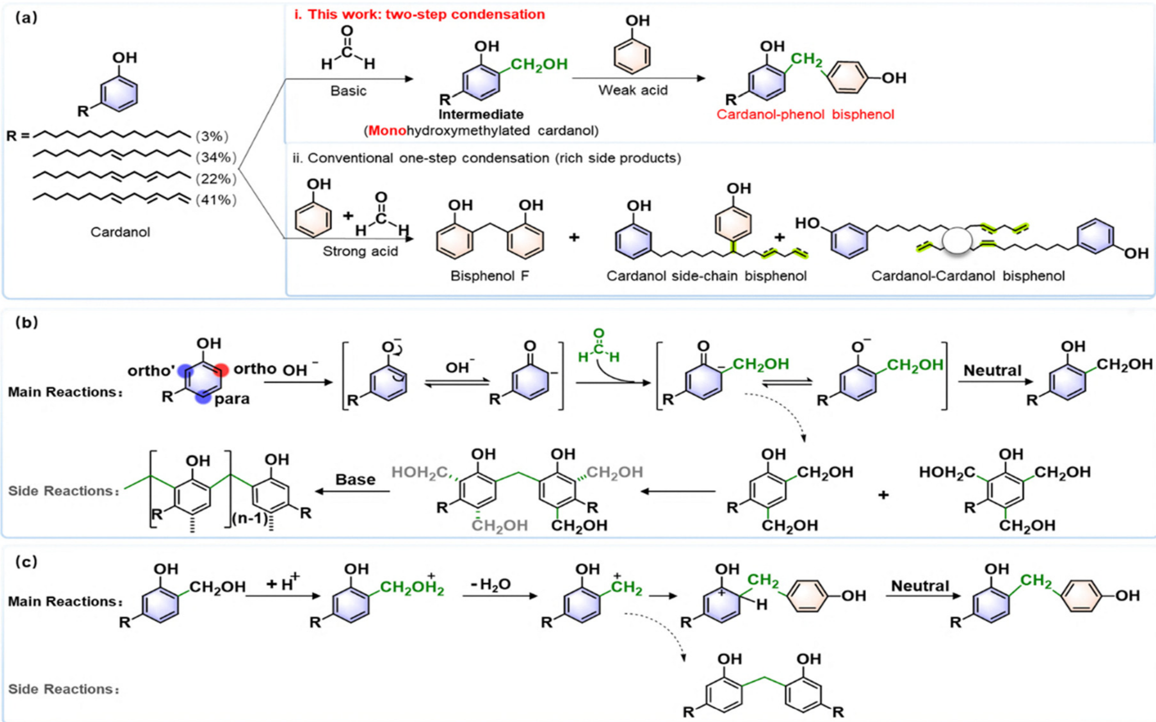

Fig. 1 (a) The condensation of cardanol and phenol by (i) a two-step condensation process using a monohydroxymethylated cardanol intermediate and (ii) the conventional one-step approach (BPF process). (b) The main and side reactions in the preparation of the monohydroxymethylated cardanol intermediate. (c) The main and side reactions in the condensation reaction of the monohydroxymethylated cardanol intermediate and phenol.

Herein, it is demonstrated that using monohydroxymethylated cardanol as the intermediate, the condensation of cardanol and phenol can be achieved *via* a two-step condensation reaction process. As shown in Fig. 1b, in the preparation of the monohydroxymethylated cardanol intermediate, the electron density of the phenolic ring increased due to the basic catalyst. The reactivity of –H at the *ortho* and *para* positions increased and it tended to react with formaldehyde to form a hydroxymethylation intermediate. Due to the steric hindrance of the long C_{15} side chain at the *meta* position, the reactivity of the *ortho*- and *para*-positions of the phenol ring was reduced, enabling controlled functionalization and yielding monohydroxymethylated cardanol. It is worth noting that under alkaline conditions, phenol is highly prone to polyhydroxymethylation, and therefore, monohydroxymethylated phenol cannot be prepared using the above-mentioned process. In this reaction, different degrees of hydroxymethylation may happen. The hydroxymethylated cardanol with different numbers of hydroxymethyl groups can react with cardanol and form large molecules with a network structure. Following the increase of the hydroxymethylated cardanol concentration, self-polymerization between different molecules may occur, forming polymeric structures. The factors influencing the side reactions in this process include reaction time, the specific composition of the raw materials, and temperature. As shown in Fig. 1c, in the condensation of the monohydroxymethylated cardanol intermediate and phenol, the hydroxymethyl group protonated under acidic conditions followed by dehydration and formation of the hydroxymethylbenzyl carbocation, which then reacted with phenol. Due to the high reactivity of the hydroxymethylbenzyl carbocation, condensation between two monohydroxymethylated cardanol intermediates may also occur and form a cardanol–cardanol bisphenol.

To minimize the side reactions in the synthesis of the monohydroxymethylated cardanol intermediate, systematic single-factor experiments have been performed to optimize the reaction parameters (SI parts 3.4). The influence of formaldehyde/cardanol ratio, reaction time, and temperature on the reaction has been investigated. The optimal conditions for attaining the highest monohydroxymethylated cardanol intermediate and minimizing byproducts were determined as the cardanol to formaldehyde molar ratio of 1 : 0.4, a reaction time of 2 hours, and a reaction temperature of 80 °C. In the condensation between the monohydroxymethylated cardanol intermediate and phenol, the influence of phenol/monohydroxymethylated cardanol intermediate ratio, reaction time, and temperature on the reaction has been studied. The optimal conditions for obtaining the highest bisphenol concentration were a molar ratio of monohydroxymethylated cardanol to phenol as 1 : 5, a reaction time of 3 hours, and a reaction temperature of 90 °C.

The reaction products were analyzed using high-performance liquid chromatography (HPLC). Fig. 2a presents the chromatogram of cardanol, which primarily exhibits four distinct peaks, labeled as 1, 2, 3, and 4, corresponding to triolefins, diolefins, monoolefins, and saturated olefins on the alkyl side chain of cardanol, respectively. Fig. 2b shows

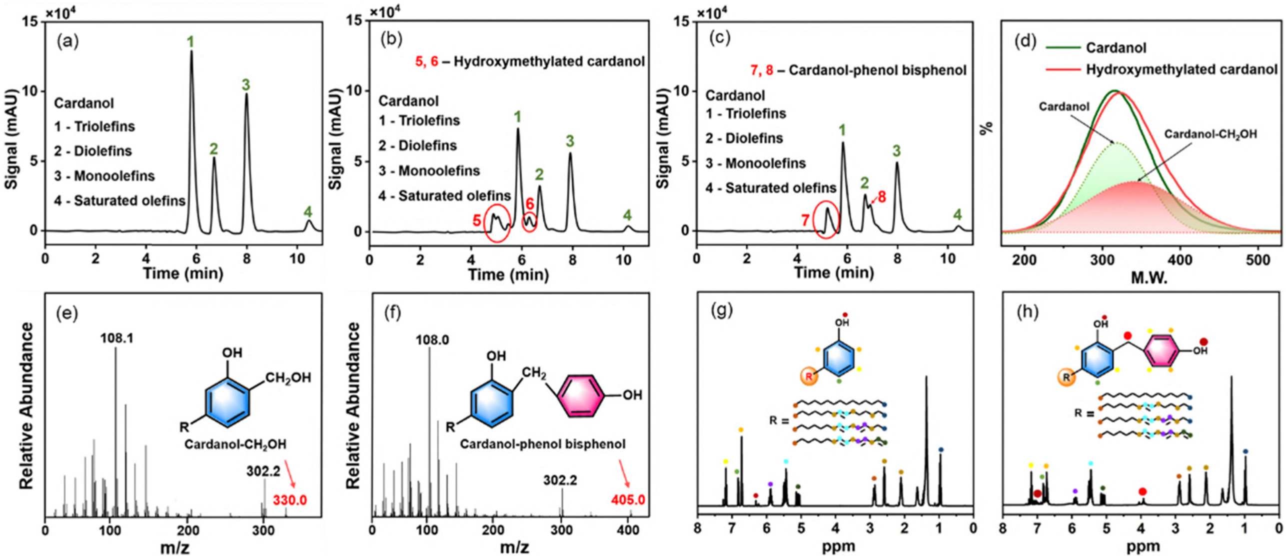

Fig. 2 High-performance liquid chromatography (HPLC) of (a) cardanol, (b) products after formaldehyde/cardanol reaction, and (c) products after monohydroxymethylated cardanol/phenol reaction. (d) The molecular weight distribution of cardanol and its products after formaldehyde/cardanol reaction. Representative mass spectra of (e) monohydroxymethylated cardanol and (f) cardanol–phenol bisphenol. ^{1}H NMR spectra of (g) cardanol, and (h) cardanol–phenol bisphenol.

the chromatogram of the products after cardanol-formaldehyde reaction under basic conditions. A significant reduction in the peak strength of cardanol was observed, attributed to the formation of hydroxymethylated cardanol through its reaction with formaldehyde under basic conditions. It should be noted that new peaks labeled as 5 and 6 occur, which show hydroxymethylated cardanol. Fig. 2c illustrates the chromatogram of the products obtained after hydroxymethylated cardanol/phenol reaction under acidic conditions. Compared to Fig. 2b, new peaks (labeled as 7 and 8) were observed, representing the synthesis of cardanol–phenol bisphenol.

Subsequent analysis using gel permeation chromatography (GPC) provided insight into the molecular weight distribution of the reaction products. Fig. 2d demonstrates that the reaction between cardanol and formaldehyde under basic conditions results in a molecular weight shift, indicating the attachment of $-CH_2OH$ groups to cardanol. The intermediate product contains 52.95% cardanol and 47.05% monohydroxymethylated cardanol. Gas chromatography-mass spectrometry (GC-MS) was then used to analyze the molecular structure of the reaction products. Fig. 2e and f display representative mass spectra of the products obtained from the cardanol/formaldehyde reaction under basic conditions and hydroxymethylated cardanol/phenol under acidic conditions, respectively. As a comparison, the GC-MS spectra of cardanol and phenol are also listed in SI part 4. Peaks at m/z values of 108 and 302 match the spectral library of cardanol and are identified as its fingerprint peaks. Peaks at m/z values of 330 and 405 are attributed to cardanol–CH_2OH and cardanol–phenol bisphenol. Nuclear magnetic resonance (NMR) spectroscopy was employed to further elucidate the molecular structure of the products as shown in Fig. 2h.[31] As a comparison, NMR spectra of cardanol were shown in Fig. 2g, respectively. Peaks corresponding to carbon and hydrogen atoms were annotated, as shown in the accompanying inner figures. Specifically, the presence of $-CH_2$ groups is confirmed by the appearance of the peak at 4.09–3.85 (ppm) in Fig. 2h. The observed changes in other peaks and the Fourier transform infrared (FT-IR) characterizations (SI part 5) are also consistent with the cardanol–phenol bisphenol structure, which further demonstrated the successful synthesis of cardanol–phenol bisphenol.

The yield of the final product was evaluated using GPC as shown in Fig. 3. The results demonstrated that the two-step reaction process produces a bisphenol compound with a yield of 40.1% (20.8% cardanol–phenol bisphenol and 19.3% cardanol–cardanol bisphenol). The residual cardanol can be further removed by molecular distillation and after the removal of

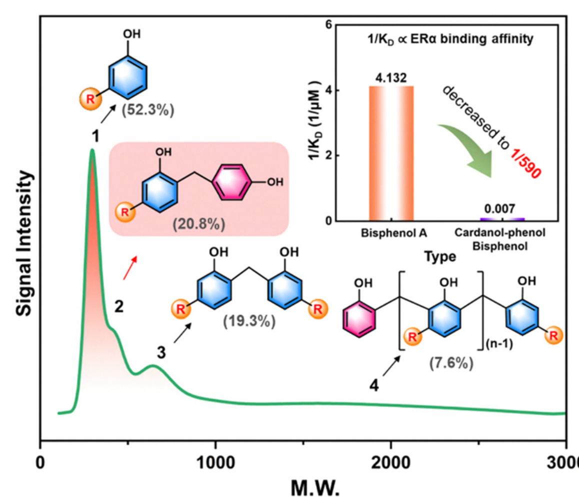

Fig. 3 The molecular weight distribution curve of the final products after the two-step reaction.

cardanol, the bisphenol product contains 43.6% cardanol–phenol bisphenol, 40.5% cardanol–cardanol bisphenol, and 15.9% cardanol–phenol polyphenol. To evaluate the strong hormone receptor binding affinity of the novel bisphenol, its binding affinity with ERα was studied using surface plasmon resonance (SPR) analysis.[32] This method enables the direct determination of binding affinity through the dissociation constant (KD), where 1/KD is proportional to the binding affinity. The results showed that the hormone receptor binding affinity of cardanol–phenol bisphenol was reduced by almost 3 orders of magnitude compared to BPA (1/590, detailed analysis in SI part 6). The low affinity between the novel cardanol–phenol bisphenol and ERα was probably attributed to its long and hydrophobic side chains.

To conclude, this paper introduces a novel two-step process to synthesize bisphenol with low hormone receptor binding affinity using cardanol, phenol, and formaldehyde. In the first step, cardanol reacts with formaldehyde to produce a mono-hydroxymethylated intermediate. This intermediate subsequently undergoes a condensation reaction with phenol, leading to the successful synthesis of a cardanol–phenol bisphenol. The resulting bisphenol exhibits several notable advantages: (i) it contains 75 wt% bio-based material, with a molecular structure featuring two benzene rings and hydroxyl groups; (ii) due to the presence of the C_{15} side chain, the molecular structure of the prepared bisphenol differs significantly from that of BPA/BPF, which decreases its binding affinity with hormone receptors by almost 3 orders; (iii) as a replacement for traditional BPA/BPF, its C_{15} side chain will help to enhance the polymer toughness, which is crucial for polymer applications. This study provides an innovative perspective and methodology for bisphenol material synthesis, showcasing significant potential for practical applications in polymer science and related fields.

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Conflicts of interest

There are no conflicts to declare.

Data availability

The data supporting this article have been included as part of the SI. See DOI: **https://doi.org/10.1039/d5cc04913j**.

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